Patents!

Economic Growth spring 2026

Lecture 3

Asger Mose Wingender

KØBENHAVNS UNIVERSITET

Why study patents?

- Romer model: crucial for growth, but also comes with drawbacks
- Controversial both inside and outside academia
- Important part of current innovation system
- Often used as proxy for innovation – including in many papers on the reading list for this course

Widely used measures of innovation

Input
- R&D spending
- Researchers

→

Output
- Patents
- New products
- Research papers

→

Effect
- TFP growth
- Firm growth

Suppose that you want to evaluate a policy aimed at fostering innovation/growth

- What metric(s) would you prefer?
- What might the drawbacks be (we discuss patents in detail later)?

Agenda

- How does the patent system work?
- Empirical tests of benefits
  - (small aside on methodology)
  - Monopoly rights: Kong and Zhao (2024)
  - Disclosure: Hegde, Herkenhoff and Zhu (2023)
- Criticisms of the patent system
- Measuring innovation using patents

The patent system

What can be patented

- Inventions, not discoveries
- Inventions must be novel and non-obvious (an *inventive step* is required)
- Inventions must have some industrial applicability
- Inventions must be of patentable subject matter
- The specifics vary across countries

Types of patents

- **On the functioning of a product**
  - Simply called “patents” in Europe
  - Called “utility patents” in the United States

- **On the design of a product**
  - “Design patents” in the United States
  - “Registered designs” in Europe (not a patent, but similar)

- **On plants**
  - In the US: Patents on plants with asexual reproduction (e.g., budding and grafting), hybrids, and GMO’s
  - In Europe: Some IP property rights for plants, but less strict that patents

Applying for a patent

- Submit application with relevant documents to patent office(s)
  - Provisional application might be submitted earlier, establishing *priority*
  - Patent families: patent applications filed in multiple countries for the same invention.

- Contents of application
  - Detailed description of the invention, including technical drawings
  - Documentation invention is novel and non-obvious compared to what has already been patented or publicly disclosed.
  - Citations of relevant older patents and scientific papers
  - Claim(s) exactly defining what is patented

- Pay fee

Patent examination

- Done by patent lawyers and technical experts (patent clerks)

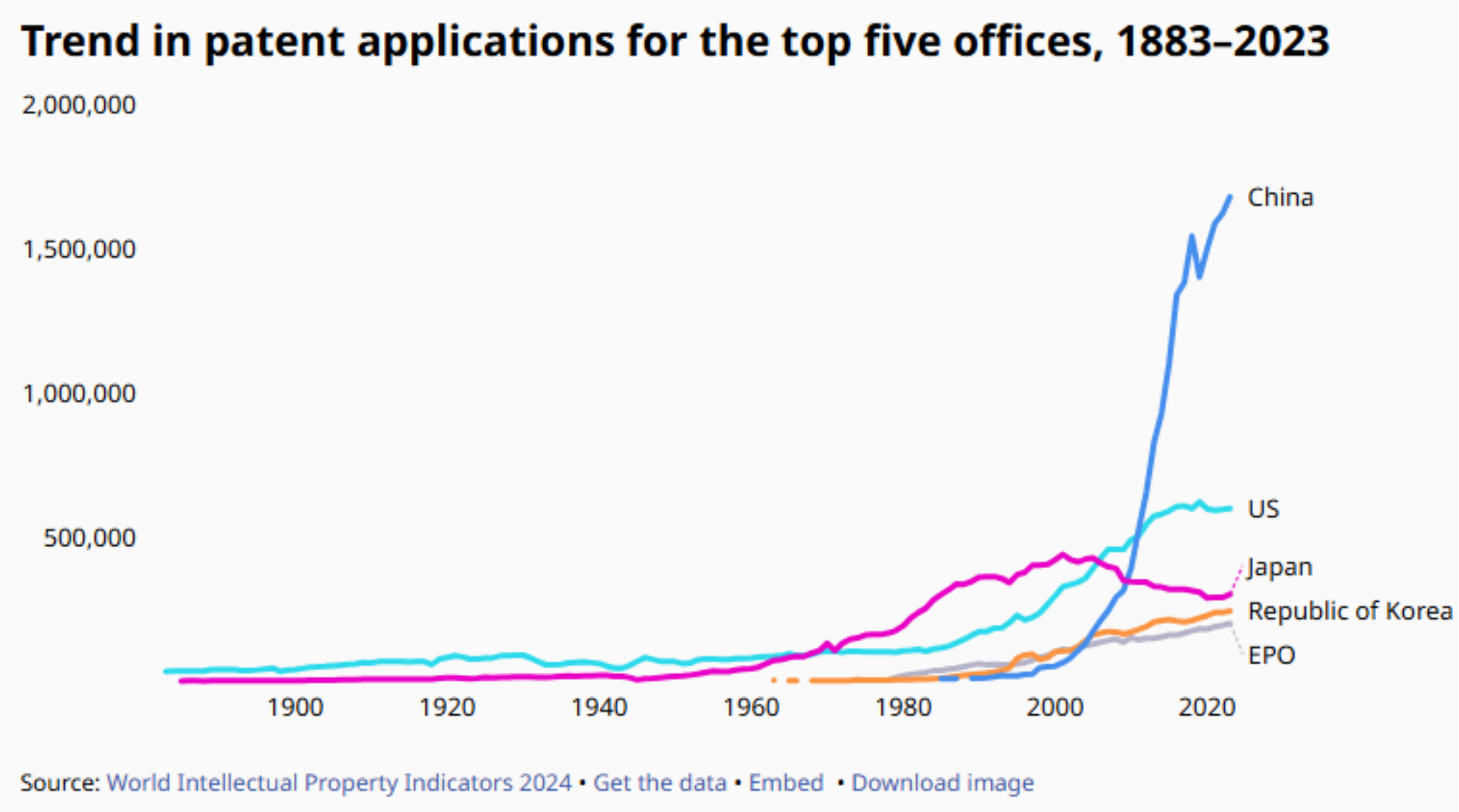
- Examination process (typically takes 2-4 years)
  - Search for “prior art”, further citations added
  - Technical and legal evaluation
  - Possible objections need to be addressed by applicants
  - Final review and decision to grant patent or not

- Application made public with a lag, typically 18 months after submission

What happens when a patent is granted?

- Gain monopoly right, typically for 20 years
- Renewal fees need to be paid periodically to maintain patent
- Can take legal action against patent violations in special patent courts

Patents by region



Empirical evidence on the effects of patents

Empirical tests of theoretical benefits of patens

Benefit 1: Incentivizes innovation (as in Romer model)

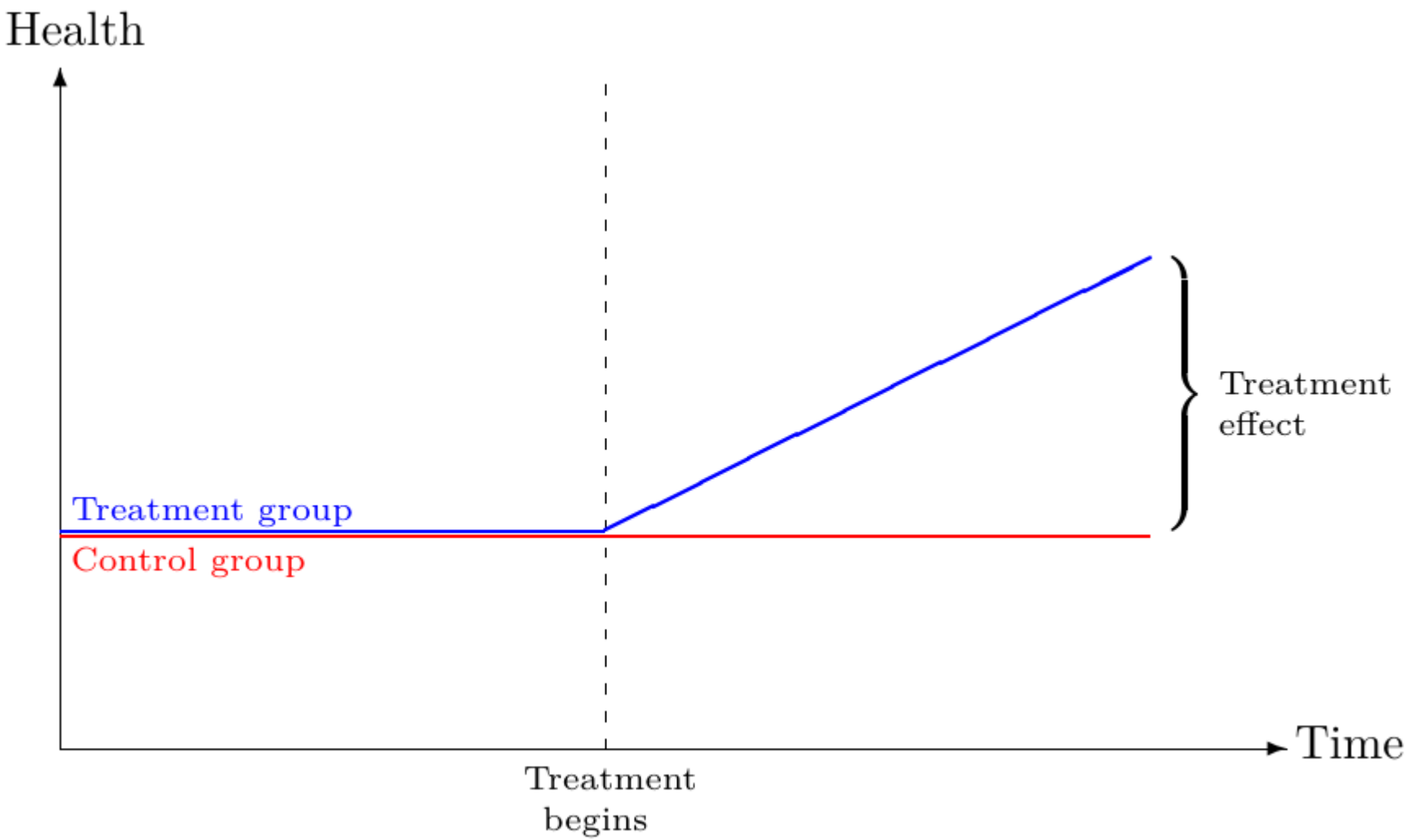
- Tested by Kong and Zhao (2024) using the GAIN act

Benefit 2: Disclosure in patent facilitates knowledge diffusion

- Tested by Hegde et al (2023) using changed regulation in the US

Both papers use *natural experiments* for causal identification

Experiments in medicine



Experiments in social science

- RCTs used for some micro-level studies
- Not feasible (or ethical) for macro studies
- Alternative: natural experiments
  - exogenous shocks affecting a parameter of interest
  - Heterogenous effects: unit of observations divided in a treatment and a control group
  - We will see many examples in this course

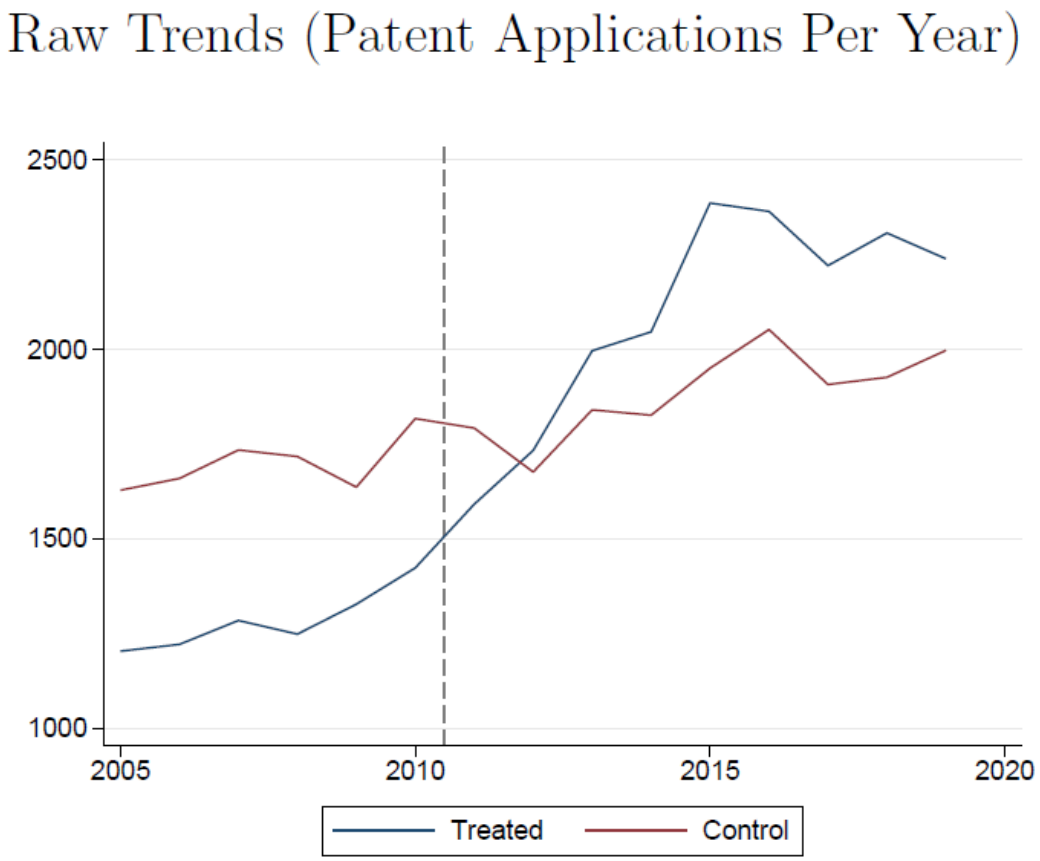
Kong and Zhao (unpublished, most recent version from 2025)

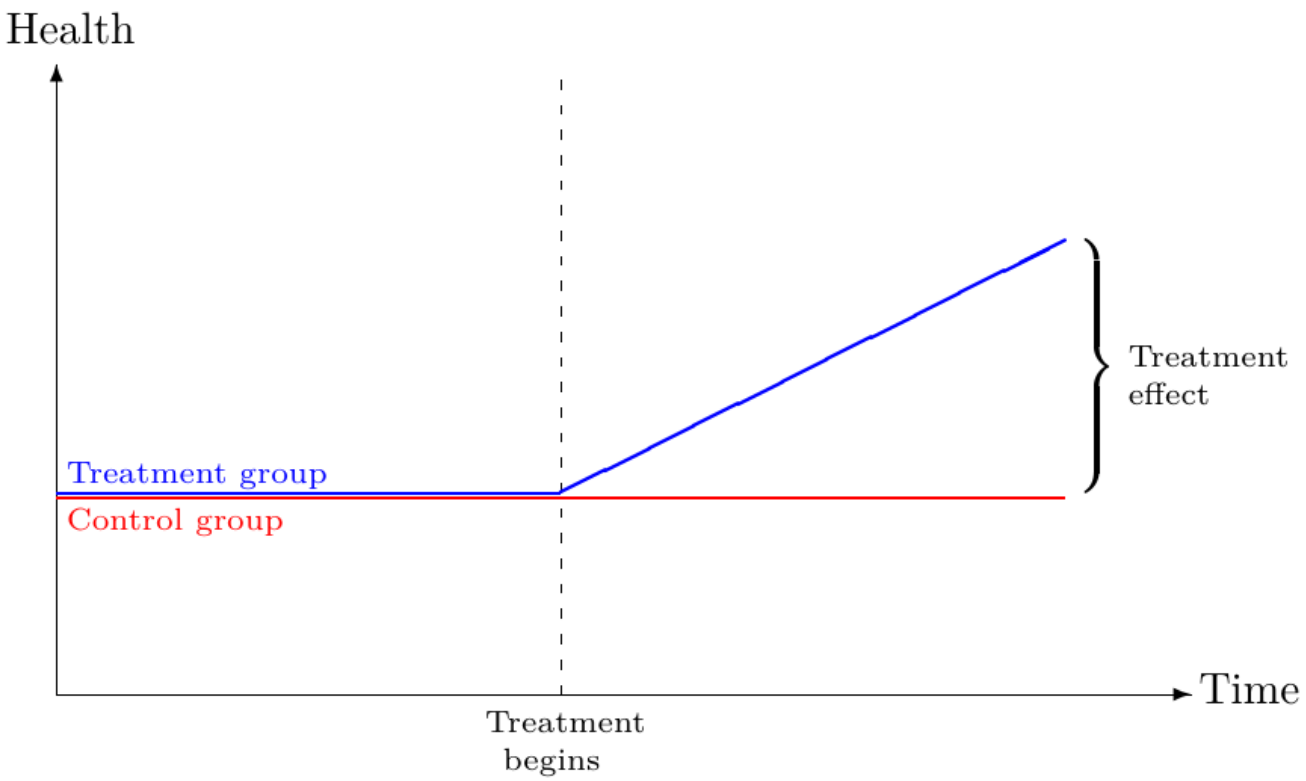
- **Background:** Medical companies abandoned R&D in antibiotics in the early 2000s, citing poor returns on investment

- **Natural experiment:** the Generating Antibiotic Incentives Now (GAIN) act
  - Five additional years of *market exclusivity* for antibiotics in the US, starting in 2010
  - Market exclusivity awarded by FDA, unrelated but similar to patent monopoly rights
  - Control group: not change for antivirals or vaccines

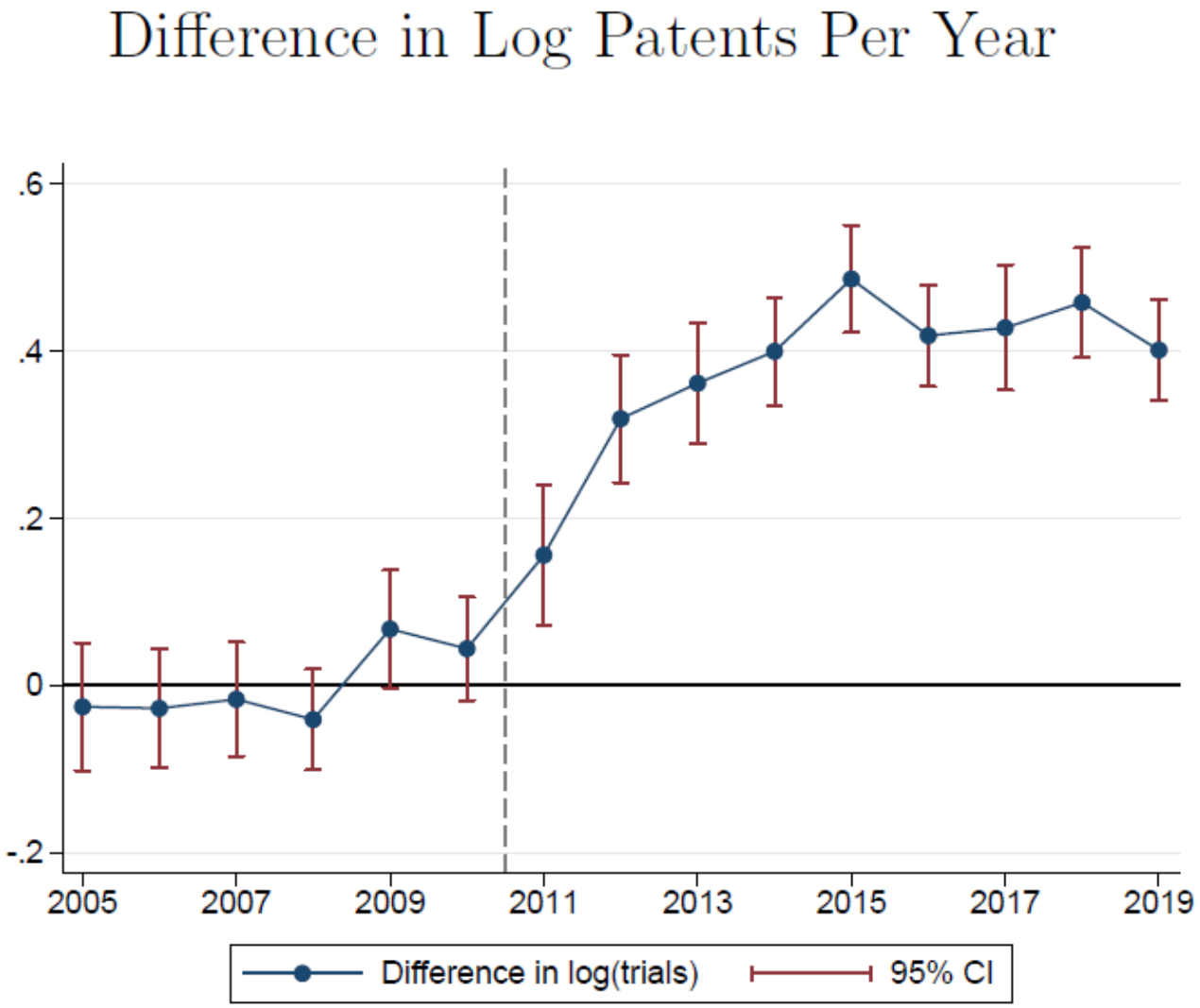
- **Outcomes:** Innovative milestones
  - Patent application, pre-clinical trials, phase 1, 2 and 3 trials, drug launches

Natural experiment in Kong and Zhao vs. stylized example





Event study analysis



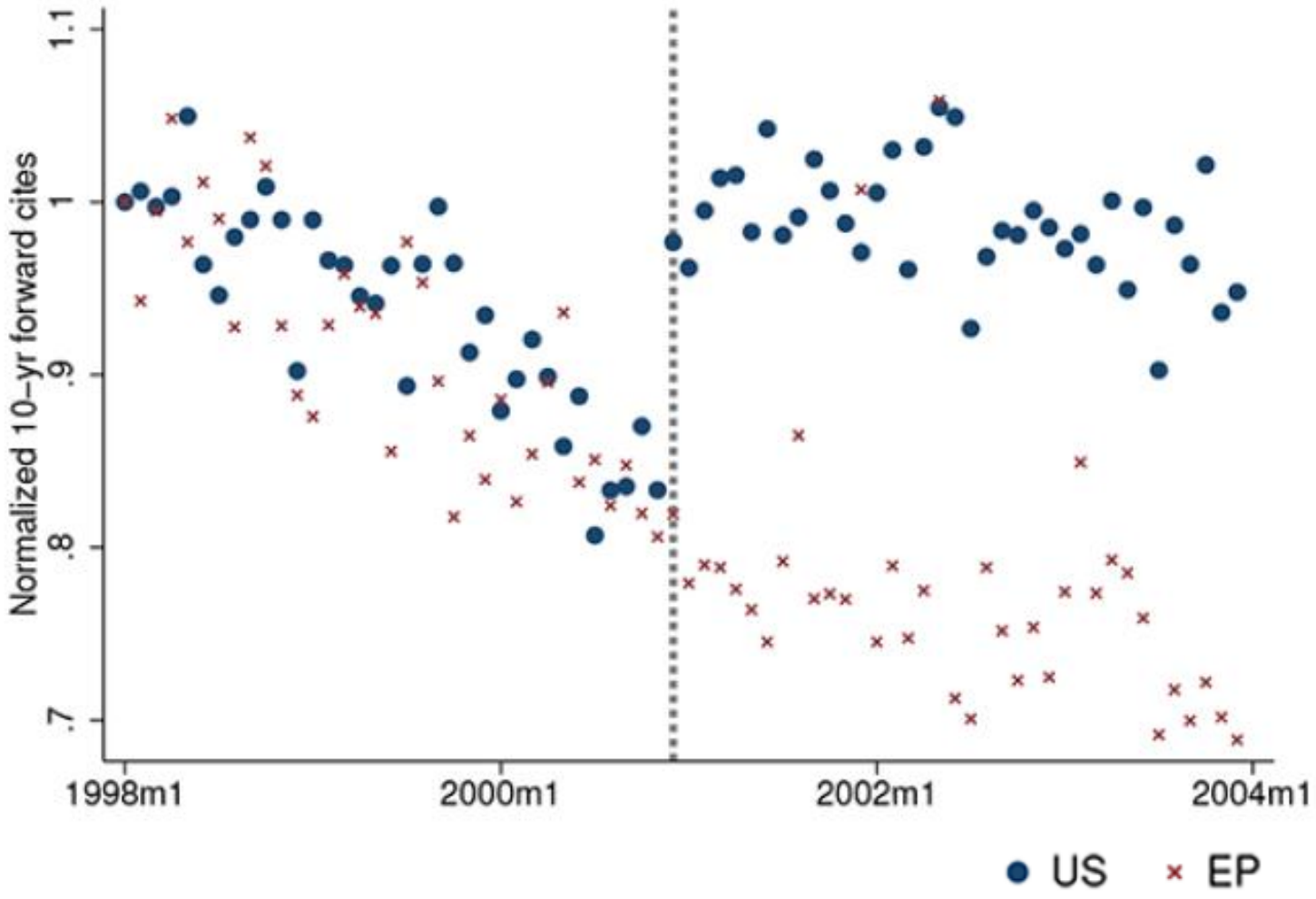
Hegde et al (2023): Testing the effect of disclosure

- **Background:** Patent applications made public in the EU after 18 months, but only when patents were granted in the US (typically after 3 years)

- **Natural experiment:** American Inventors Protection Act (AIPA) of 1999 aligned US system with the EU.
  - Compare innovations patented in both the US and the EU
  - US patents are "treated" after 1999, with EU patents acting as a control group

- **Measure of knowledge diffusion:** forward citations
  - How many patents cites the patent X years after disclosure?

Forward citations of twin patents



From Hegde et al (2023), Fig. 7, panel D.

Some concerns

Might results reflect that people just cite US patent instead of EU twin?

- Adjustment for EU patents citing US patents, and not EU twin
- We should see EU patent citations drop after AIPA, but they do not

Why do citations of US patent increase if EU equivalent already out there?

- Paper argues for search costs, language barriers and other frictions
- Could there be other explanations?

What can we conclude about disclosure and innovation?

Criticisms of the patent system

(and patents as a measure of innovation)

Theoretical pros and cons

Pros

- Incentivizes innovation (as in Romer model)
- Innovation disclosed in patent ⇒ facilitates knowledge diffusion

Cons

- Monopolies are bad
- High costs of litigation may stifle follow-up innovation

Empirical criticism

The case against patents can be summarized briefly: there is no empirical evidence that they serve to increase innovation and productivity, unless productivity is identified with the number of patents awarded—which, as evidence shows, has no correlation with measured productivity.

Boldrin and Levine 2013

- **Main point:** If we find no benefits of patents in data, the cons are likely more important
- **Also:** Strong criticism of patents as indicator for innovation
  - Does the criticism apply to the papers by Hong and Zhao and Hegde et al?
  - We get back to patents as indicators for innovation later

Econ history perspective (Moser, 2016)

- We cannot study the total effect of the current patent system due to lack of a control group, but history offers some lessons (more in the paper)

- Many important inventions not patented
- **World's fairs:** no difference in showcased innovations across countries with and without patent systems
- **But:** patents important when secrecy is hard, and modern methods of reverse engineering have made secrecy harder.

Patents help incumbents and patent trolls

- Patent litigation costly in terms of time and money
- Incumbents keep out upstart competitors by taking out vaguely formulated patents, which can be used as a basis for legal harassment.
- Patent trolls do the same, hoping to extract money from actual innovators

US00D670286S

(12) **United States Design Patent**
Akana et al.

(10) **Patent No.: US D670,286 S**
(45) **Date of Patent: ** Nov. 6, 2012**

(54) **PORTABLE DISPLAY DEVICE**

(75) Inventors: **Jody Akana**, San Francisco, CA (US); **Bartley K. Andre**, Menlo Park, CA (US); **Daniel J. Coster**, San Francisco, CA (US); **Daniele De Iuliis**, San Francisco, CA (US); **Evans Hankey**, San Francisco, CA (US); **Richard P. Howarth**, San Francisco, CA (US); **Jonathan P. Ive**, San Francisco, CA (US); **Steve Jobs**, Palo Alto, CA (US); **Duncan Robert Kerr**, San Francisco, CA (US); **Shin Nishibori**, Portola Valley, CA (US); **Matthew Dean Rohrbach**, San Francisco, CA (US); **Peter Russell-Clarke**, San Francisco, CA (US); **Christopher J. Stringer**, Woodside, CA (US); **Eugene Antony Whang**, San Francisco, CA (US); **Rico Zorkendorfer**, San Francisco, CA (US)

(73) Assignee: **Apple Inc.**, Cupertino, CA (US)

(**) Term: **14 Years**

(21) Appl. No.: **29/379,722**

(22) Filed: **Nov. 23, 2010**

Related U.S. Application Data

(63) Continuation of application No. 29/354,599, filed on Jan. 26, 2010, and a continuation of application No. 29/353,307, filed on Jan. 6, 2010, now Pat. No. Des. 627,777.

(51) **LOC (9) Cl.** **14-02**

(52) **U.S. Cl.** **D14/341**

(58) **Field of Classification Search** ..... D14/341–347, D14/137, 138, 138.1, 138 A, 138 AA, 138 R, D14/138 AB, 138 AC, 138 AD, 138 C, 138 G, D14/496, 203.1, 203.3, 203.4, 203.5, 203.7, D14/426, 427, 432, 429, 129, 130, 420, 439–441, D14/448, 125, 147, 156, 218, 247–248, 250, D14/389, 388, 315–318; D10/65, 104; D18/6–7; D21/324, 329, 330; D19/26, 59, 60; 455/556.1, 455/556.2, 566, 575.1, 90.3; 379/433.04, 379/433.01, 433.06, 916; 345/169, 173, 345/901, 905; 361/679.26, 679.27, 679.3, 679.55, 679.56

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2,424,630 A 7/1947 Perez

(Continued)

FOREIGN PATENT DOCUMENTS

AU 315078 7/2007

(Continued)

OTHER PUBLICATIONS

Ballmer, S., Spiegel Online, "Bill Gates, Tablet PC: "Mira" war 2002 die große Vision des Microsoft-Gründers—kam aber wohl zu früh für den Markt," (http://www.spiegel.de/netzwelt/gadgets/bild-670472-46770.html), product announced 2002, 4 pages.

(Continued)

Primary Examiner — Barbara Fox
(74) *Attorney, Agent, or Firm* — Sterne, Kessler, Goldstein & Fox PLLC

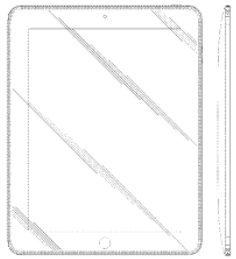
(57) **CLAIM**

The ornamental design for a portable display device, as shown and described.

DESCRIPTION

FIG. **1** is a bottom front perspective view of a portable display device showing our new design;
FIG. **2** is a bottom rear perspective view thereof;
FIG. **3** is a front view thereof;
FIG. **4** is a rear view thereof;
FIG. **5** is a top plan view thereof;
FIG. **6** is a left side view thereof;
FIG. **7** is a right side view thereof; and,
FIG. **8** is a bottom plan view thereof.
The shade lines in the Figures show contour and not surface ornamentation.
The broken lines in the Figures show portions of the portable display device which form no part of the claimed design.

1 Claim, 5 Drawing Sheets



Political economy criticism

- Who benefits from increasing the scope of patent system?
- Who loses from increasing the scope of patent system?
- Which of the two groups are likely to wield the most influence in politics?

Should we abolish the patent system?

"If we did not have a patent system, it would be irresponsible, on the basis of our present knowledge of its economic consequences, to recommend instituting one. But since we have had a patent system for a long time, it would be irresponsible, on the basis of our present knowledge, to recommend abolishing it."

Fritz Machlup (1958), quoted in Boldrin and Levine (2014)

What does a better patent system look like

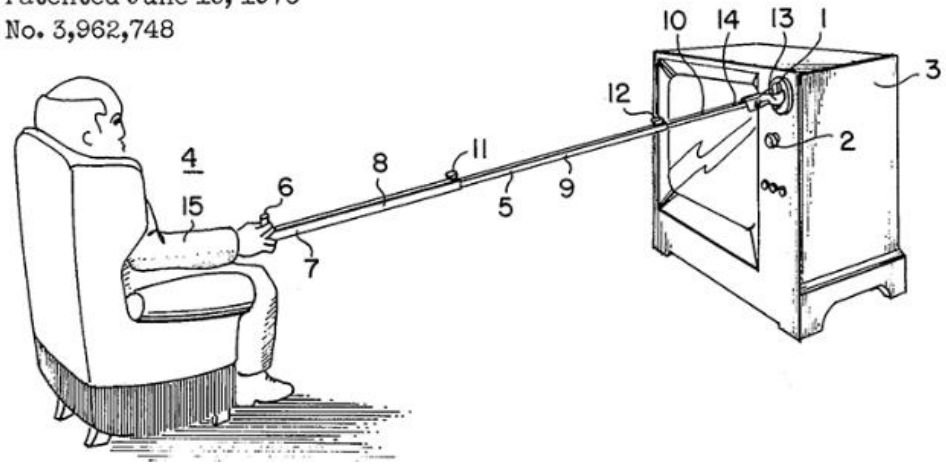
Selected recommendations from Boldrin and Levine (2013)

- Reduce the set of what can be patented
- Shift the legal and judicial balance from patent holders to innovators
- Tailor the length and scope of patents to different sectorial needs
- Grant patents not just based on novelty, but also on economic grounds (i.e., take both costs and benefits into account)
- …

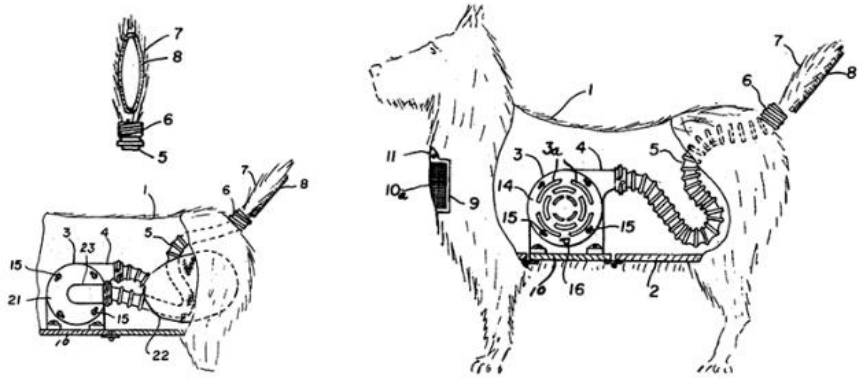
Patents as a measure of innovation

Many useless patents

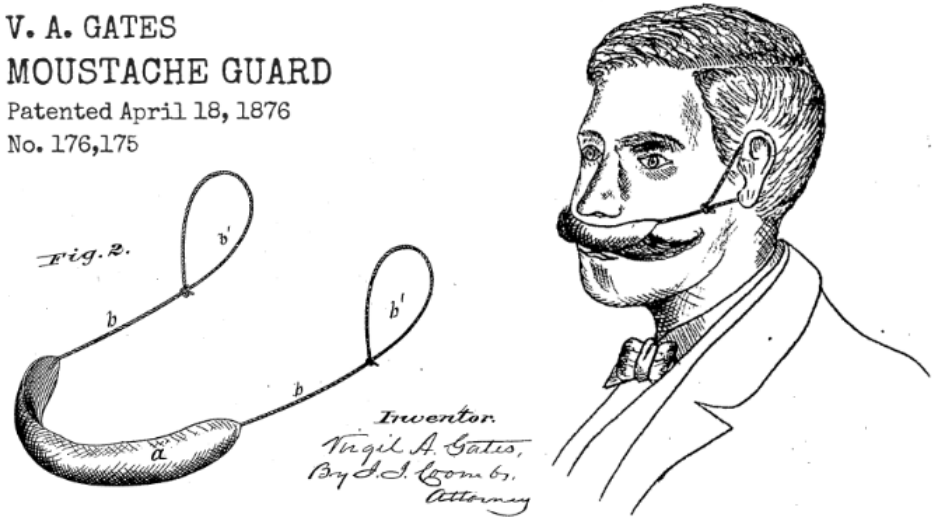
CHRIS T. MICHAELS
TV CONTROL DEVICE
Patented June 15, 1976
No. 3,962,748



ANNE MARGARET ZALESKI
COMBINATION TOY DOG AND VACUUM CLEANER
Patented November 13, 1973
No. US 3,771,192



V. A. GATES
MOUSTACHE GUARD
Patented April 18, 1876
No. 176,175



Source: WIPO's annual list of weird patents

Arguments against patents as innovation indicator

- Many innovations are not useful
  - Counterpoint: Many are, and useful ones can often be identified in data
- Many innovations are not patented
  - Counterpoint: Many are
- Regulation drives patenting, no info in time series and cross-country variation
  - Counterpoint: for a given regulatory system, more patents might mean more innovation
- Little evidence that patent systems increases TFP growth etc.
  - Counterpoint: TFP and economic growth driven by other things than just patented innovation, so link might be hard to identify

- **In sum:** patents are noisy proxies for innovation, use with caution!

Benefits of using patents in research

Contain lots of useful information:

- Application date
- Name and address of inventor and owner
- Technology class
- Description of the innovation
- Citations of earlier, related patents
- Citations of related academic literature

Measuring patent quality

Traditional methods

- Forward citations (how many patents cite the patent)
- Forward citations *relative* to backward citations
- Patent renewal
- Patent family size

New methods using AI and language models

- Determining novelty and similarity based on textual analysis
- More sophisticated ways of studying citation patterns

Where can I find patent data?

- Actual patents available online, but hard to use
- Aggregate data on patent numbers from WIPO, OECD, and patent offices
- USPTO has a data set with citations, technology class etc. for all US patents
- Google patents has everything, but is a bit complicated to use